

MIMO Techniques in 5/6G (F-js851-3*)

Technical Milestone Report

Kaiyuan Bao (kb784)
Homerton College, University of Cambridge

Supervised by Prof. Albert Guillen i Fabregas and Alexander Hamilton (Nokia)

January 3, 2026

Abstract

This report investigates

.
. .
. .
. .
. .
. .
. .

1 Introduction

Mobile telecommunications have experienced rapid development over the past two decades. Starting from the deployment of third-generation (3G) systems in the early 2000s, followed by the introduction of fourth-generation Long-Term Evolution (4G LTE) in the early 2010s, each generation has delivered substantial improvements in data rate and user experience. For end users, the most noticeable change has been the transition from limited and slow web browsing to smooth video streaming and real-time online services. ///These improvements were enabled by key physical-layer innovations in 4G, including orthogonal frequency division multiple access (OFDMA), advanced channel coding schemes, and the use of multiple antennas.///

Building upon the capabilities of 4G, the fifth-generation mobile communication system (5G) has been designed to further increase data rates, reduce latency, and support a wide range of services. Three main usage scenarios are defined for 5G: enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable and low-latency communications (URLLC) [1]./// To meet these diverse requirements, multiple-input multiple-output (MIMO) techniques play a central role by exploiting spatial degrees of freedom to improve spectral efficiency and link reliability. In particular, large-scale MIMO systems enable significant throughput gains through spatial multiplexing and beamforming.///

The Third Generation Partnership Project (3GPP) is a global standardisation body established in 1998 to develop specifications for mobile communication systems. Its scope has expanded from third-generation systems to include the standardisation and continuous evolution of 4G, 5G, and emerging technologies beyond 5G. ///Today, 3GPP defines the technical specifications that ensure interoperability and global consistency across mobile communication networks, including detailed physical-layer procedures for MIMO transmission and channel state information (CSI) feedback.///

This research project aims to provide insights that can support future 3GPP standardisation discussions related to MIMO techniques in 5G and beyond. Simulations are conducted in MATLAB using a framework aligned with 3GPP specifications. This report introduces the relevant background concepts, presents the adopted simulation methodology, and summarises the current progress and preliminary findings of the project.

2 Background and Technical Overview

The downlink PDSCH; CSI, time granularity: period , spatial granularity: code book type, frequency granularity: wide/subband; HARQ

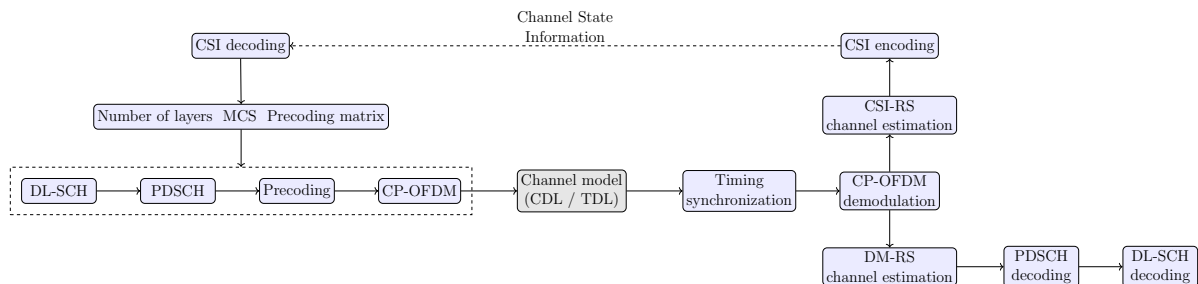


Figure 1: Overview of the CSI-based downlink transmission and feedback framework, adapted from [2]

/// As illustrated in Figure 1, the downlink transmission relies on channel state information (CSI) obtained through reference signal measurements at the receiver. The user equipment (UE) estimates the downlink channel based on CSI reference signals (CSI-RS). Based on the estimated channel, the CSI is encoded into different feedback components, including the Channel Quality Indicator (CQI), Precoding Matrix Indicator (PMI), and Rank Indicator (RI). These parameters are fed back to the transmitter to support link adaptation and precoding decisions. At the transmitter, the reported CSI is used to configure key downlink transmission parameters, such as the modulation and coding scheme, the number of transmission layers, and the precoding matrix applied to the data symbols.

As the CSI feedback is reported periodically, the accuracy of the transmitted CSI may degrade over time in time-varying channels, leading to outdated channel knowledge at the transmitter.

///

3 Simulation Framework

Table 1 summarises the baseline system and channel configuration adopted in this study, which is aligned with the assumptions commonly used in 3GPP evaluation studies, including TR 38.753 [3]. ///Low- and high-mobility scenarios are considered. For high-mobility cases, a vehicular speed of 120 km/h is assumed, corresponding to a maximum Doppler shift of approximately 390 Hz for an FR1 mid-band deployment. ///

///The CDL-C channel model is selected as it represents a typical non-line-of-sight urban macro environment and is widely used in 3GPP evaluation studies. A 4×4 MIMO configuration with up to four transmission layers is considered, reflecting a representative FR1 deployment while maintaining a manageable simulation complexity. This choice

Table 1: System and channel configuration used in the simulations

Parameter	Value
Carrier frequency band	FR1 mid-band
Subcarrier spacing	30 kHz
Channel bandwidth	106 RBs
Antenna configuration	4 Tx, 4 Rx
Maximum number of layers	4
Channel model	CDL-C
Mobility scenarios	Low (10 Hz), High
Simulation type	Downlink link-level

enables the effects of CSI feedback parameters to be examined without introducing additional variability associated with very large antenna arrays. ///

This study focuses on investigating the impact of CSI feedback parameters on downlink transmission performance. The baseline CSI configuration follows the reporting principles and parameter ranges specified in relevant 3GPP specifications, including TS 38.101-4 [4]. Both the baseline configuration and the investigated parameter ranges are summarised in Table 2.

///Unless otherwise stated, all simulations use the baseline CSI and HARQ configuration listed in Table 2. In each experiment, only a subset of the CSI parameters under investigation is varied, while the remaining parameters are kept fixed at their baseline values to ensure a controlled comparison.///

Table 2: CSI feedback configuration

Parameter	Baseline setting	Investigated values
CSI report periodicity	10 slots	2, 5, 10, 20, 40 slots
CQI reporting mode	Wideband	Wideband / Subband
PMI reporting mode	Wideband	Wideband / Subband
Subband size	16 RBs	4, 8, 16, 32 RBs
RI configuration	Adaptive	Fixed / Adaptive
Codebook type	Type I	Type I
Codebook mode	Mode 1	Mode 1 / Mode 2

4 Model Validation and Simulation Observations

To validate the accuracy of the developed simulation framework, a comparison is conducted against results reported by other firms in R4-2509413 [5] under the baseline parameter settings from the previous section. ///AAV 3 vs 1Y///

As shown in Figure 2, the proposed simulation exhibits a similar throughput trend to the firm-standard baseline. The SNR deviation at the 30% and 70% throughput points remains within a worst-case margin of 4.1 dB and 2.2 dB, respectively, indicating good agreement between the results.

///Minor discrepancies are observed, which can be attributed to differences in implementation assumptions and modelling details. In particular, the reference results implicitly include impairment margins reflecting practical UE considerations, whereas the

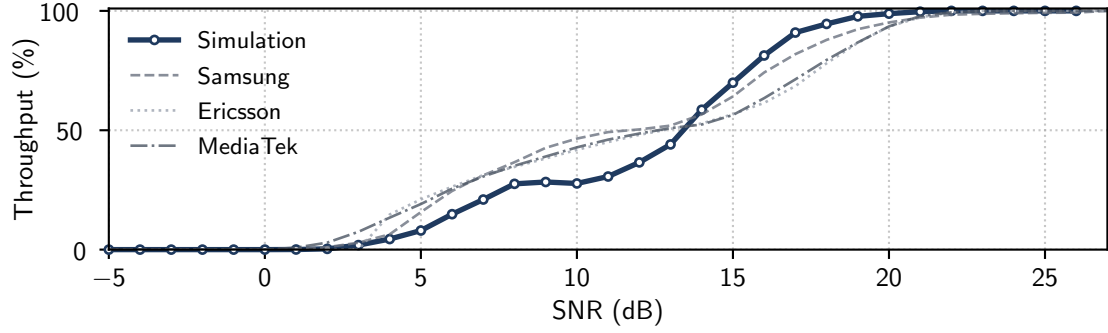


Figure 2: Throughput comparison between the simulation and the firm-standard baseline

present simulation focuses on a pure link-level evaluation. ///

Before investigating the CSI feedback parameters, it is useful to examine the impact of HARQ on the throughput and BLER. Figure 3 compares the throughput and BLER results with HARQ enabled and disabled.

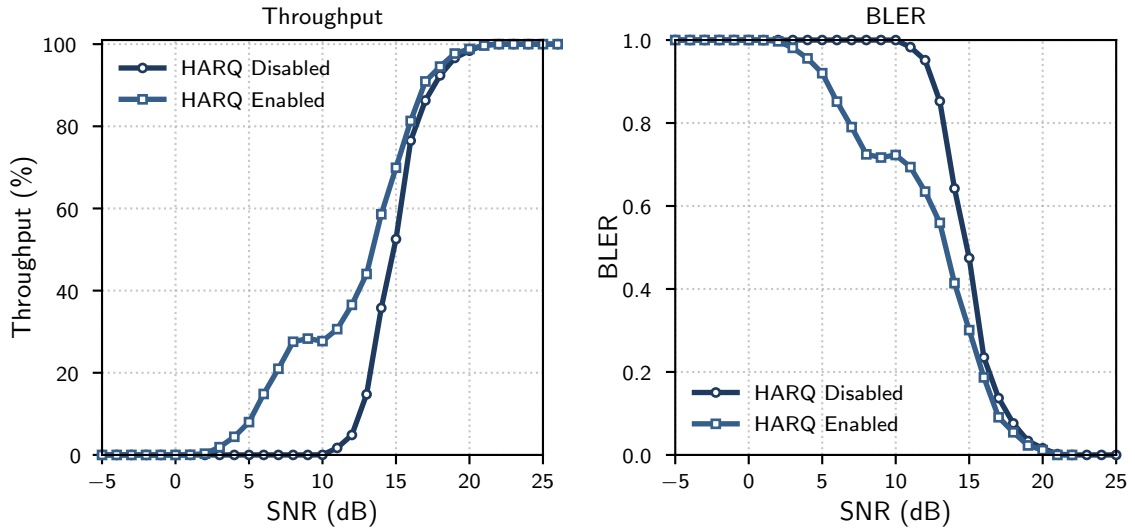


Figure 3: Throughput and BLER performance comparison with HARQ enabled and disabled

With HARQ enabled, a significant reduction in BLER and improvement in the throughput are observed, particularly in the low- to medium-SNR region less than 15dB. When the SNR is beyond 20dB, whether HARQ is on or off has no impact on the overall performance. All of the transmission success in one go, and thus no re-transmission is required. /// These observations motivate the subsequent analysis, where HARQ is selectively enabled or disabled to isolate the impact of CSI feedback parameters. ///

To further examine the behaviour of high-rank transmission, the performance of an eight-layer transmission is analysed. Figure /// illustrates the throughput performance of individual codewords under the considered transmission setup. ///

///The results show that the throughput performance of the two codewords differs across the SNR range, reflecting the unequal channel conditions experienced by different spatial layers. This behaviour is consistent with the underlying MIMO transmission

structure and highlights the importance of CSI feedback in supporting high-rank spatial multiplexing.///

5 Summary and Next Steps

6 References

- [1] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic press, 2020.
- [2] MathWorks, “NR PDSCH Throughput Using Channel State Information Feedback,” 5G Toolbox Documentation, 2025, online; accessed on 2025-12-24.
- [3] 3GPP, “Study on spatial channel model for demodulation performance requirements for NR,” 3GPP, techreport Release 19, Sep. 2025.
- [4] 3GPP, “User Equipment (UE) radio transmission and reception. Part 4: Performance requirements,” 3rd Generation Partnership Project (3GPP), techreport TS 38.101-4, Sep. 2025, Release 19.
- [5] 3GPP, “Results collection table for NR SCM,” 3rd Generation Partnership Project (3GPP), Bengaluru, India, resreport R4-2509413, Aug. 2025, RAN4 Meeting #116.